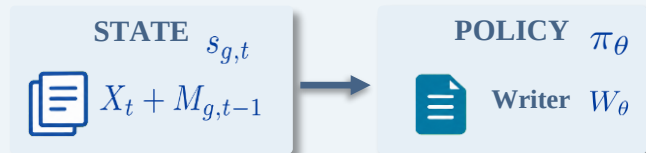


A ON-POLICY ROLLOUT

independent candidate states

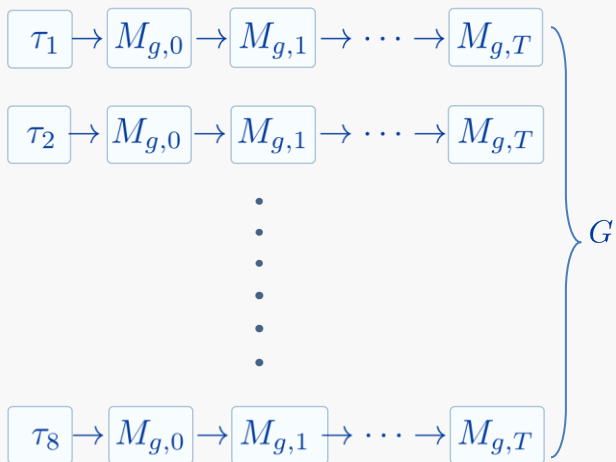


ADD

UPDATE

NOOP

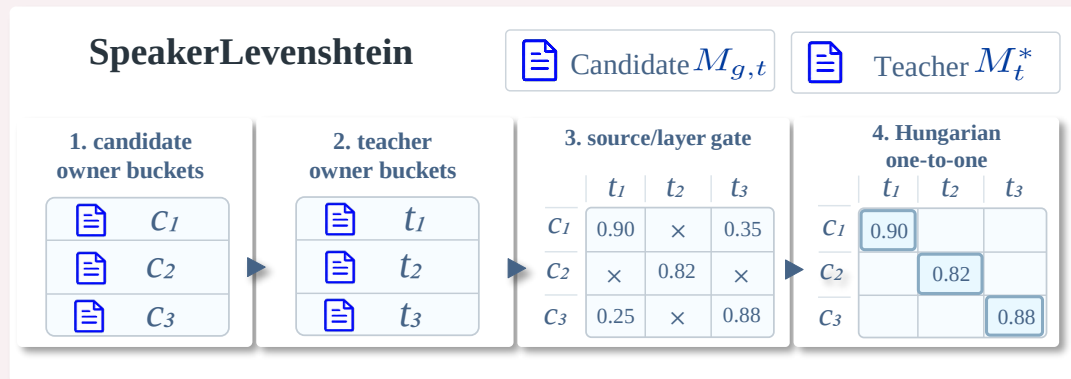
Independent Rollouts



B DUAL CREDIT

dense local structure + delayed downstream utility

LOCAL STRUCTURE at each (g, t)



$$R_{state} = w_1 Score_{avg} + w_2 Score_{worst}$$

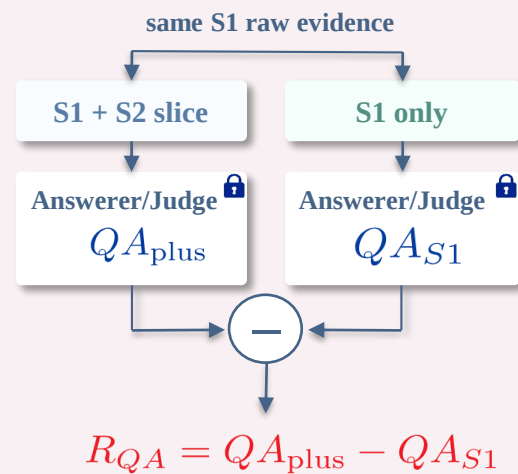
$$R_{mem} = w_s R_{state} + w_c R_{chain}$$

Backfill Credit $\gamma^{(T-1)} R_{QA}$



$$r_{g,t} = w_v R_{valid} + w_m R_{mem} + w_{qa} \gamma^{(T-1-t)} R_{QA} + P_{struct}$$

TERMINAL QA after $t = T - 1$



C POSITION-WISE GRPO UPDATE

compare candidates only at the same transition position

